

EARLY GRADE MATHEMATICS ASSESSMENT

EGMA Report

NOVEMBER 2022



FEDERAL MINISTRY OF EDUCATION
ABUJA NIGERIA.

**FEDERAL MINISTRY OF EDUCATION ABUJA
NIGERIA**

**PILOT
EARLY GRADE MATHEMATICS ASSESSMENT (EGMA)**

Report

November 2022

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ACRONYMS

B.Ed.	Bachelor of Education
B.Sc.	Bachelor of Science
CM&MLA	Curriculum Monitoring & Monitoring of Learning Achievement
EGRA	Early Grade Reading Assessment
EGMA	Early Grade Mathematics Assessment
EPR&D	Educational Planning Research and Development
M.Sc.	Master of Science
NCE	National Certificate In Education
P2	Primary 2
P3	Primary 3
P4	Primary 4
PGDE	Post Graduate Diploma in Education
WASH	Water Sanitation and Hygiene
ECCDE	Early Childhood Care & Development Education
LSI	Letter Sound Identification
ALPP	Accelerated Learning Program PLUS
CESLY	Core Education Skills for Liberian Youth (USAID project)
DIF	Differential item functioning
EdData II	Education Data for Decision Making (USAID project)
MANOVA	Multivariate analysis of variance
NCTM	National Council of Teachers of Mathematics
TIMSS	Trends in International Mathematics and Science Study
USAID	United States Agency for International Development

EXECUTIVE SUMMARY

The Early Grade Mathematics Assessment (EGMA) is designed to measure the foundational knowledge of basic mathematics skills in lower primary classes. The assessment is sectioned into five sub-task sections that are each designed to ascertain a particular aspect of numeracy. These subtasks include number identification (Number Identification), reasoning about magnitude (Number Discrimination), recognition of number patterns (Missing Number), addition and subtraction (Addition and Subtraction, Levels 1 and 2), and Arithmetic stories (Word Problems).

The assessment was done by Primary 2 and Primary 3 learners who had never participated in any previous assessments of this nature. The assessment involves the participation of a total of 540 learners in 18 schools drawn from the six geo-political zones of the country at three (3) schools per state.

The report is sectioned into three parts; the first is the introduction and methodology of the study, the second showed the result of the study by subtask and the final section is the summary of findings, recommendations and conclusion.

The result from the study showed that there is generally a weak knowledge of mathematics based on learners' performance across subtests while the percentages of correct responses was also considered very low in subtests where learners made good attempts at providing responses.

External factors such as the reported low availability of WASH facilities and inadequate classroom furniture was also observed. This could result in low attendance rate which also contribute to the performance of the learners.

To sufficiently address the low performance of learners in reading mathematics, there is need for a holistic review of the methodologies and curriculum used in teaching or introducing learners to mathematics. Adequate training on new teaching methodologies should be provided to teachers so as to better educate learners and adopt effective technics in in teaching the subject. Other factors such as provision of teaching aids, counters and age- appropriate classroom furniture will improve the result and learning experience.

1 BACKGROUND

Several studies over the years have shown that learners with a strong foundation in mathematics or arithmetic during the early grades remain a key factor to future success in mathematics, which is instrumental in the development of workplace skills and knowledge (Malloy, 2008; Nunes & Bryant, 1996; Steen, 2001; U.S. Department of Education, 2008). In addition, basic mathematical reasoning is key to everyday activities such as shopping and personal finance. Recent meta-analysis also suggest that early mathematics skills predict later reading skills just as much as early reading skills (Duncan et al., 2007; Romano et al., 2010). There is a growing recognition among policy makers, donors, and educators, of the importance of making evidence-based policy and programming decisions. However, Ministries of Education in developing countries and donor organizations are challenged by a lack of solid information about student learning outcomes in mathematics, particularly in the early grades. Building on the success of, and great demand for, the Early Grade Reading Assessment (EGRA), which was developed in 2006 and implemented to date in more than 50 countries, the United States Agency for International Development (USAID) contracted with RTI International in 2008 to develop an assessment of early grade mathematics competencies. The result was the Early Grade Mathematics Assessment (EGMA), an orally administered assessment of the core mathematical competencies taught in primary grades 1. The EGMA, to date, has been used in 14 countries around the world. Please note that the original EGMA was revised in 2011 and renamed the Core EGMA.

The Core EGMA is an assessment of early mathematics learning, with an emphasis on number and operations. The Core EGMA consists of six subtests (referred to as tasks in the instrument) that, taken together, can produce a snapshot of children's knowledge of the competencies that are fundamental in early grade mathematics. These competencies (and their respective subtest titles) include number identification (Number Identification), reasoning about magnitude (Number Discrimination), recognition of number patterns (Missing Number), addition and subtraction (Addition and Subtraction, Levels 1 and 2), and Arithmetic stories (Word Problems). The Core EGMA is an oral assessment and individually administered to students by trained assessors. Many mathematics assessments require children to be able to read in order to solve problems. Because the Core EGMA is designed for the early grades, which is when children are just beginning to learn how to read, the oral administration does not confound a child's ability to read or write with a child's ability to do mathematics.

An instrument such as the Core EGMA holds significant promise in providing stakeholders, from Ministries of Education to aid agencies, with the information essential to making informed local and national system changes in teacher education and curriculum development and implementation. Accurate and efficient assessment of early grade mathematical skills does not provide all the information necessary to effect change for the better in mathematics teaching and learning. However, adding an instrument that is valid across national borders and in multiple languages can provide the Ministries of Education with an overall assessment of students' abilities based on what is known about children's mathematical development,

independent of regional curricula. Depending on the level of analyses, this information can be tailored to examine specific within-country geographical or political sub-regions. The results of these analyses can give Ministries of Education the ability to target interventions, such as teacher training, on specific topics or sub-regions. It is important to understand that students' scores are more than just a measure of their abilities in a specific skill—they are also a measure of the students' environment. Myriad factors can affect student learning, including socioeconomic status, nutrition, skills and education of teachers, instructional time, parental education, insurgency, threats or localized violence, and family responsibilities.

OVERALL OBJECTIVES

The curriculum Monitoring and Monitoring of Learning Achievement (CM&MLA) unit of the Federal Ministry of Education is responsible for monitoring the effectiveness of the learning curriculum and methodology in early childhood education and perform periodic reviews in accordance with the results of their assessment. One of the assessments is the Early Grade Mathematics Assessment (EGMA). The objectives of the pilot EGMA study is to;

1. Adapt existing EGMA instrument and perform a comprehensive review in accordance with curriculum of primary 2 and primary 3 learners.
2. Perform a baseline assessment in selected states across the country of which results will inform the planning of a broader study in all states.
3. Provide early insight of the current state of mathematics education in early childhood education.

2 METHODOLOGY

2.1 Sample

The assessment was carried out in six (6) states of the federation (these includes Abia(SE), Delta(SS), Gombe(NE), Jigawa(NW), Nasarawa(NC), and Ondo(SW)), therefore selecting one state from each of the six (6) geopolitical zones in the country. In each state a sample of three (3) schools was selected randomly from list of available schools provided by the State Universal Basic Education Board (SUBEB). The selection reflects one school each from the three senatorial districts of the selected states making a total of 18 schools in the study. At the school level, a random sample of 15 learners was drawn from multiple classes enrolled and present on the day of visiting the school at both Primary 2 and Primary 3 levels. The result of which ensures that 30 learners per school and 540 across the six geo-political zones of the country were selected for the pilot study.

2.2 Assessment

Based on the standard guidelines for the study, the pilot assessment covered the provisions recommended for the Core Early Grade Mathematics Assessment (EGMA) only. Skills tested in the assessment are based on the prescribed five(5) subtasks as highlighted in Table 2.1 below:

Table 2.1: Assessment Content

Skill evaluated	Number of items	Duration
Number identification	20 numbers	60 seconds
Number discrimination	10 pair of numbers	Untimed task
Missing number identification	10 items	Untimed task
Addition	10 problems	60 seconds
Subtraction	10 problems	60 seconds
Word problems	6 problems	Untimed task

2.3 Learner`s interview

Based on empirical evidence from several studies that multiple factors contribute to learners' performance in school that are often traceable to home and societal influences, especially community participation in education based on family educational background, a learners' specific interview was administered to collect information on learners' home environment and other demographic variables (such as age, learners' perception about school and education, learner's household possession of selected items and facilities, etc.). The result of the interviews were collated to provide additional insight into the learners performance.

2.4 Assessment/Interview Administration

In order to align with the global best practices in the Administration of the EGMA, both the assessment and learners' interview was administered using tangerine application installed on mobile devices. Assessors provided learners with a stimulus requesting the child to read to the hearing of the assessor who marks the responses on the mobile device as either correct or wrong. For some subtasks, learners were provided with counters, pen, and paper to use in solving the problems.

3 ANALYSIS OF RESULTS

This section analyses results from each subtask disaggregated by State. Specifically, for each subtask state and national averages (computed as average scores from across the six (6) geo-political zones) were computed to allow for comparison. Results from this analysis are presented in Table 3.1.

Table 3.1: Average Scores by State and Subtask

(1) States	(2) Number Identification	(3) Number Discrimination	(4) Missing Number	(5) Addition	(6) Subtraction	(7) Arithmetic Story
Abia	22.7	6.4	3.5	8.6	5.9	1.7
Delta	27.0	7.0	3.3	9.3	5.5	1.7
Gombe	17.4	6.6	5.3	7.9	6.4	4.0
Jigawa	14.4	4.3	2.2	5.4	3.5	1.1
Nasarawa	18.2	5.9	2.9	7.3	4.8	3.0
Ondo	23.5	6.1	2.8	7.4	4.4	1.0
National	20.5	6.0	3.3	7.6	5.0	2.1

To provide more insights into the results, additional analysis of the data was carried out where scores from each subtask were grouped into different categories to compute the percentage of learners by category. The analysis was then rounded up by combining results from the six states to create a National average as reported and presented in Tables 3.2 through Table 3.7.

3.1 Number Identification Subtask

The first subtask in the EGMA study was number identification. Learners were given stimuli containing 20 numbers, ordered randomly and required to point and say the name of each number within 60 seconds. Column (2) in Table 3.1 shows results from number identification subtask by state and the

average across the six zones (national average). As reflected in the table, the national average score for number identification was approximately 21 numbers per minute. Comparing the states, learners from Abia state had the highest average of 22.7 numbers per minute with the lowest performance of 14.4 numbers per minute for learners in Jigawa state. Most learners have scores within the range of 16-35 with 23% in P2 and 24% in P3 which is followed by the 6-15 score range with 13% in both classes. The overall average score for this subtask is 17.3 for P2 and 23.5 for P3.

Table 3.2: Score of Number Identification Subtask by State

Score	State						National (%)
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	
0	1.05	0.00	0.00	6.45	2.06	0.00	1.62
1-5	3.16	4.44	13.64	23.66	21.65	11.83	13.13
6-15	24.21	13.33	36.36	34.41	22.68	23.66	25.72
16-35	58.95	56.67	45.45	30.11	39.18	50.54	46.76
36+	12.63	25.56	4.55	5.38	14.43	13.98	12.77

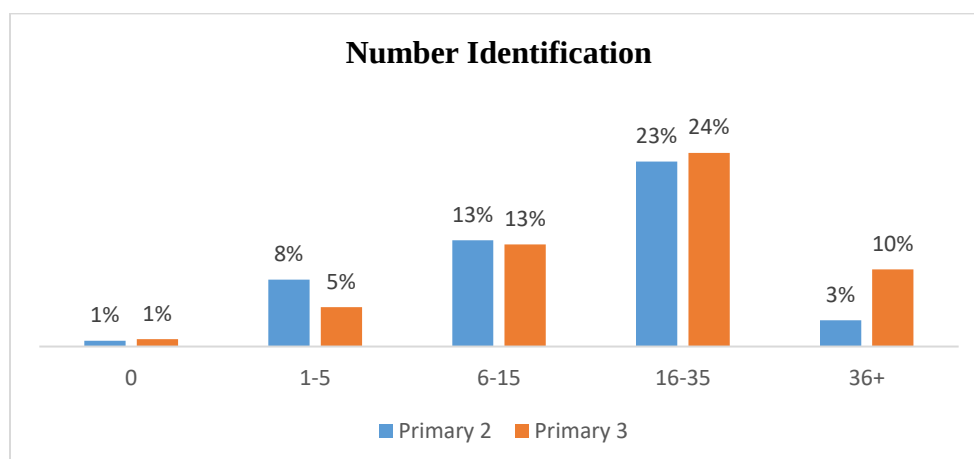


Figure 3.1: Number Identification Score by Grade

3.2 Number Discrimination Subtask

Learners were given 10 pairs of numbers and required to point and say the greater of the two numbers. The subtask was untimed and had 10 items, with the maximum scorable points of 10. Average scores for this subtask for each state is provided in column (3) of Table 3.1. Analyzing the results for this subtask, it can be observed that the national average was 6, with learners from Delta having the highest average of 7. Conversely, learners from Jigawa state recorded the lowest average of 4.3. The percentage of student that scored over 7+ in this subtask in 20% for P2 and 31% for P3. The overall average score for this subtask is 5.4 and 6.6 for P2 and P3 learners respectively.

Table 3.3: Score of Number Discrimination Subtask by State

Score	State						
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	National(%)
0	4.21	3.33	6.82	22.58	9.28	11.83	9.71
1-3	9.47	12.22	11.36	18.28	13.40	15.05	13.31
4-6	36.84	25.56	19.32	25.81	29.90	18.28	26.08
7+	49.47	58.89	62.50	33.33	47.42	54.84	50.90

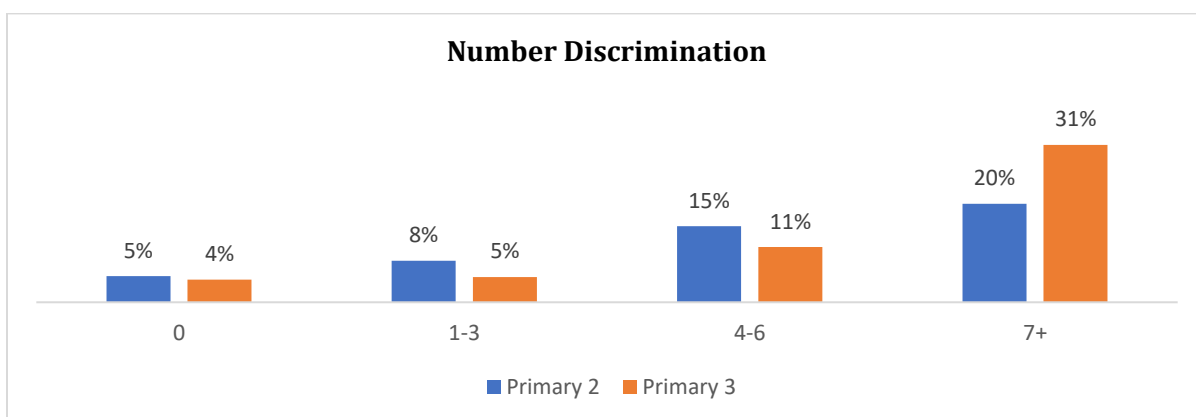


Figure 3.2: Number Discrimination Score by Grade

3.3 Missing Number Identification Subtask

Missing number identification was another skill tested during the assessment. This subtask involved identifying by reading, the pattern of available numbers from a set of 4 numbers with one missing number. Results for this subtask are presented in Column (4) of Table 3.1 disaggregated by state. Comparing the states, it can be deduced that learners from Gombe state had the best performance, averaging 5.3 marks out of 10. In contrast, learners from Jigawa state had the least performance with an average of 2.2 marks out of 10. At the national average, the score for this subtask stood at 3.3 out of 10 marks. More than half of the learners performed poorly scoring within the range of 1-3 marks for this subtask (26% for P2 and 24% for P3). The average score in this subtask is 2.9 out of 10 marks for P2 and 3.7 out of 10 marks for P3 respectively

Table 3.4: Score of Missing Number Subtask by State

Score	State						
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	National(%)
0	4.21	7.78	0.00	21.51	10.31	19.35	10.61
1-3	55.79	48.89	29.55	59.14	57.73	46.24	49.82
4-6	26.32	37.78	40.91	13.98	25.77	23.66	27.88
7+	13.68	5.56	29.55	5.38	6.19	10.75	11.69

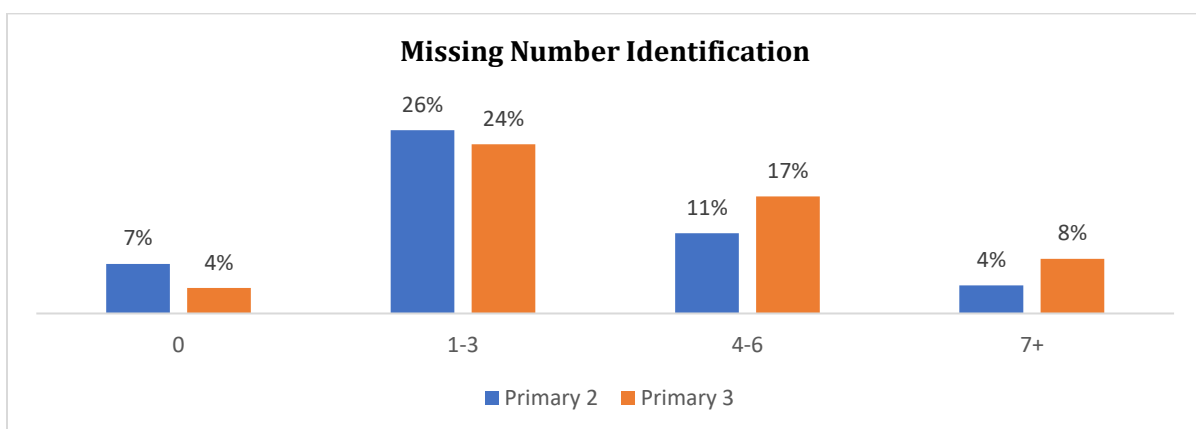


Figure 3.3: Missing Number Identification Score by Grade

3.4 Addition and Subtraction Subtasks

Learners were given addition and subtraction problems to solve. In each case, the test was designed to increase in difficulty and timed for 60 seconds. Columns (5) and (6) of in Table 3.1 contain results from addition and subtraction subtasks respectively. Comparing results for the two subtasks, it is obvious that learners across the states performed far better in solving the addition problems compared to the subtraction problems. Precisely, learners' average score in the addition subtask for the entire sample is approximately 8 addition problems per minute as against a far lower average of 5 problems per minute. Looking at state level scores, learners from Delta state had the highest average score of 9.3 addition problems per minute. in contrast, learners from Jigawa states had the lowest average score of 5.4 additions per minute. Results of subtraction subtask indicate that Abia state had the best performance, averaging 5.9 correct subtractions per minute. Again, scores of learners in Jigawa state averaged 3.5 correct subtractions per minute. In the addition subtask, learners had an average score of 6.7 in P2 and 8.5 in P3. While in the subtraction subtask, the average score is 4.4 and 5.7 for P2 and P3 learners respectively.

Table 3.5: Score of Addition Subtask by State

Score	State						
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	National(%)
0	2.11	4.44	7.95	22.58	3.09	12.90	8.81
1-5	25.26	25.56	18.18	37.63	39.18	35.48	30.40
6-10	50.53	37.78	53.41	24.73	37.11	25.81	38.13
11-15	12.63	20.00	17.05	9.68	11.34	12.90	13.85
16+	9.47	12.22	3.41	5.38	9.28	12.90	8.81

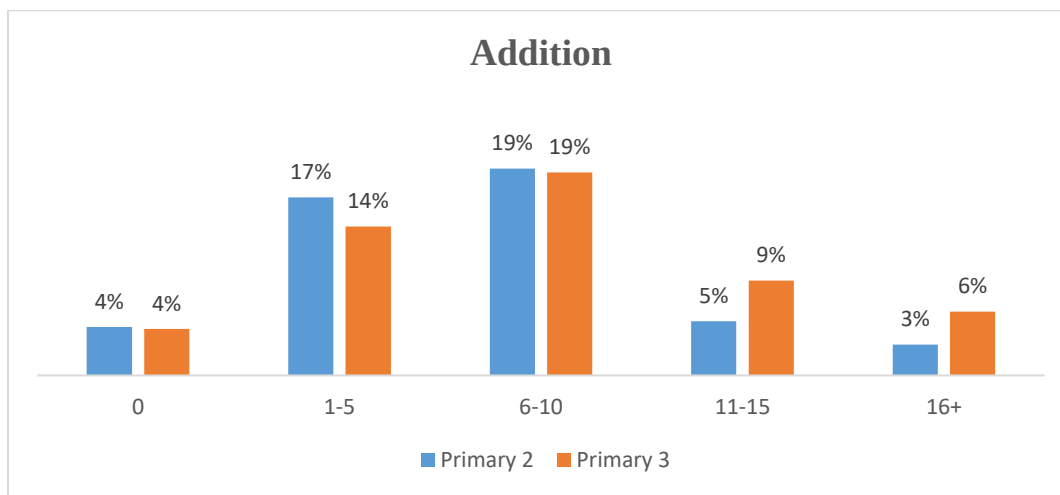


Figure 3.4: Addition Subtask Score by Grade

Table 3.6: Score of Subtraction Subtask by State

Score	State						
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	National(%)
0	12.63	20.00	10.23	30.11	13.40	30.11	19.42
1-5	49.47	32.22	32.95	40.86	46.39	34.41	39.57
6-10	26.32	35.56	50.00	25.81	29.90	26.88	32.19
11-15	2.11	8.89	4.55	3.23	8.25	6.45	5.58
16+	9.47	3.33	2.27	0.00	2.06	2.15	3.24

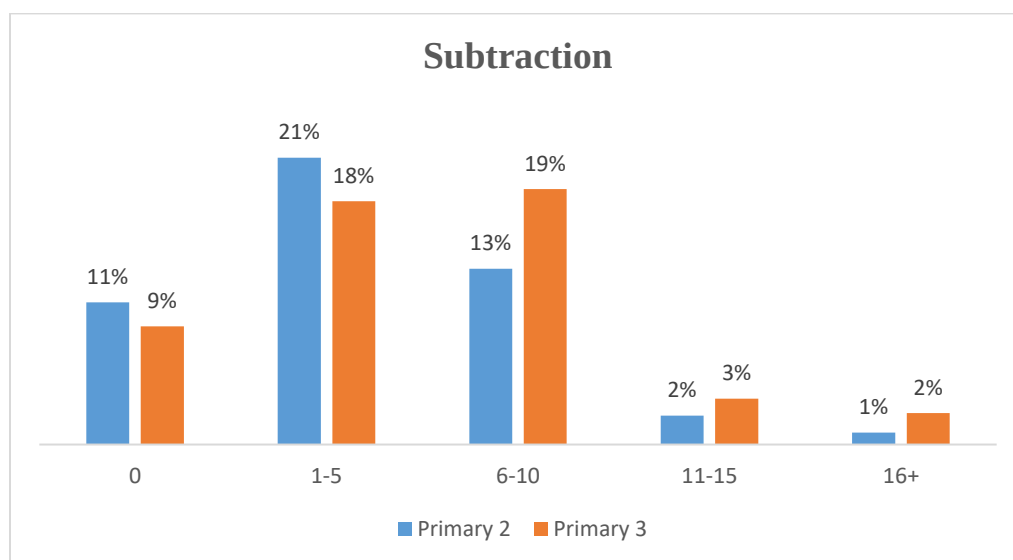


Figure 3.5: Subtraction Subtask Score by Grade

3.5 Arithmetic Story (Word Problems) Subtasks

The last subtask in EGMA involved arithmetic word problem. An assessor read to the hearing of a learner a simple arithmetic problem and requested that the child answer, on which basis, the assessor marked the learner's response as either correct or wrong. The subtask had a total of six problems and therefore the maximum possible points were 6. Column (7) of Table 3.1 shows that the national average score in arithmetic story subtask was approximately 2 out of 6, with Gombe states having the highest score of an average of 4 marks. For this subtask, learners in Ondo state had the least average score of 1.0, falling slightly behind Jigawa state with an average of 1.1. The highest percentage of learners (20% for P2 and 23% for P3) scored within the range of 1-3. The overall average score in this subtask is 1.9 for P2 and 2.3 for P3 respectively

Table 3.7: Score of Arithmetic Story (Word Problem) Subtask by State

Score	State						
	Abia(%)	Delta(%)	Gombe(%)	Jigawa(%)	Nasarawa(%)	Ondo(%)	National(%)
0	30.53	30.00	7.95	54.84	11.34	54.84	31.65
1-3	53.68	60.00	25.00	34.41	49.48	37.63	43.53
4+	15.79	10.00	67.05	10.75	39.18	7.53	24.82

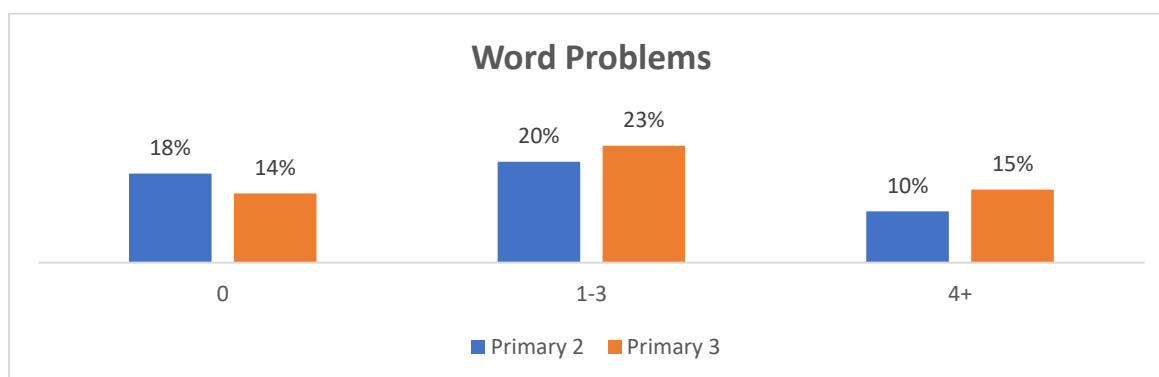


Figure 3.6: Word Problem Subtask Score by Grade

4 RESULTS OF LEARNERS INTERVIEW

This section provides a brief highlight of selected learners attributes. Variables analyzed in this section include learners' age, socioeconomic status, attendance of nursery class before enrolment into Primary 1, and absenteeism. Results are presented in Table 4.1.

According to Table 4.1, majority of the learners interviewed/assessed fall within the age bracket of 6 to 8 years (46.0 percent) P2. followed by age group of 9-11 years (P3). This age category accounts for 45.86 per cent of the learners interviewed. This result indicates that the learners are slightly behind grade-appropriate age of 7(Pry 2) and 8(primary 3) in the expected year of the basic education schooling Calendar. Delays in school enrolment, household conditions, distance to school locations etc. are some of the possible reasons that may be responsible for this disparity.

To measure the socioeconomic status of each child interviewed, questions were asked on child's household basic possession and access (Such as Radio/television, access to toilets/latrines and other factors on a scale of 14 items/facilities. For each item or facility, a score of 1 was assigned if available/or accessed by the household and 0 if otherwise. A tabulation of the results on pupil's socioeconomic status, showed a majority the learners, that is, 47.48 percent of the learners, came from households where 4-6 of the items/facilities listed are available. The extreme of that indicated that only 1.08 of the learners were reported to have a score of zero on

socioeconomic status. This is compared with 5.22 percent of the learners whose households possess 11-14 items/facilities.

Attendance of Pre-Primary /nursery classes (Early Childhood Care Development-ECCD) now introduced in majority of schools across the nation could have had significant impact on learning achievement. Information was gathered on this attribute during learners' interview. Results emanating from the analysis of these variables indicate that slightly more than half of the learners interviewed (approximately 52 percent) attend nursery/ ECCD classes before enrolling into Primary I.

Table 4.1: Pupil characteristics

Variable/Question	Response/Answer	Percentage
Age Category	3-5	1.44
	6-8	46.04
	9-11	45.86
	12+	6.65
Socioeconomic Status Score	0	1.08
	1-3	23.56
	4-6	47.48
	7-9	22.66
	10+	5.22
Gender	Boy	46.94
	Girl	53.06
Grade	Primary 2	48.20
	Primary 3	51.80
Have you attended nursery before primary 1?	No	48.02
	Yes	51.98
Do you have a mathematics book at school?	No	33.99
	Yes	66.01
Did you eat today before coming to school?	No	6.83
	Yes	93.17

Do you think school is important?	No	5.04
	Yes	94.96
Were you absent from school any day last week?	No	69.42
	Yes	30.58

Pupil's perception about schooling is important for learning. The children were asked about their opinion on the importance or not of school in general. The responses from this question were quite encouraging with around 95 of the learners assessed indicating that they feel school is important for children.

Pupil absenteeism has a lot of negative implications on learning achievement. Analysis of responses from question on whether a child was absent from the school anytime last week reveal that a significant proportion of the learners, accounting for around 31 percent, were absent from school in the week preceding the interview/assessment. This calls for concern from all stakeholders.

5 RESULT OF TEACHER'S INTERVIEW

The quality of teachers and learning environment have long been established as a good indicator to measure learners' performance in school. Teachers were interviewed to establish this assertion and the analysis of data gathered from teacher interviews are presented in this section. Variables analyzed include teacher gender, their qualifications, experience, employment status, and grade level.

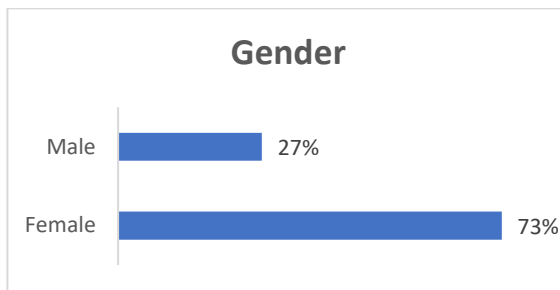


Figure 5.1: Teacher Gender Distribution

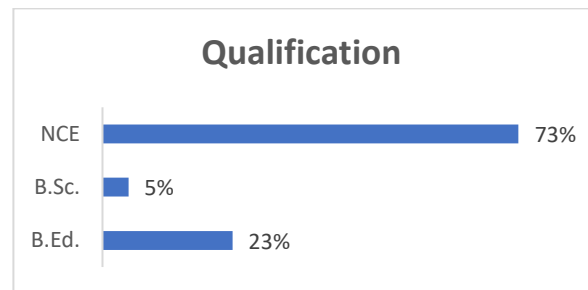


Figure 5.2: Teacher Qualification

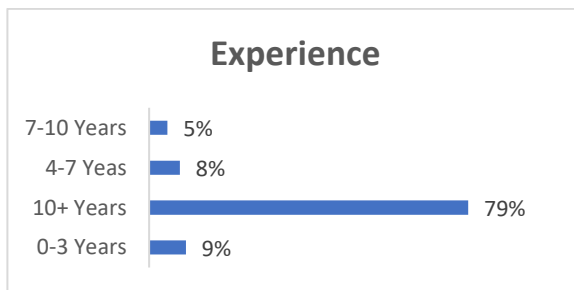


Figure 5.3: Teacher Years of Experience

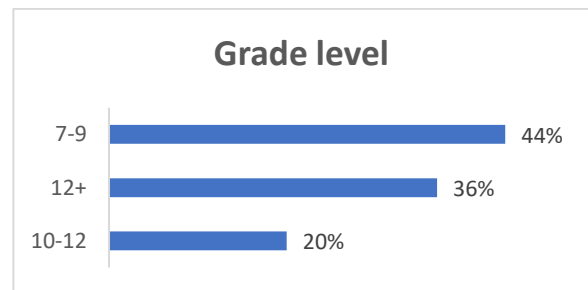


Figure 5.4: Grade level

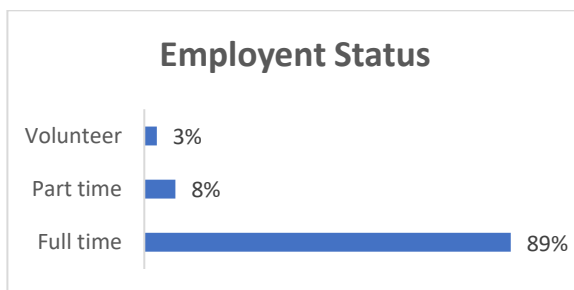


Figure 5.5: Teachers Employment Status

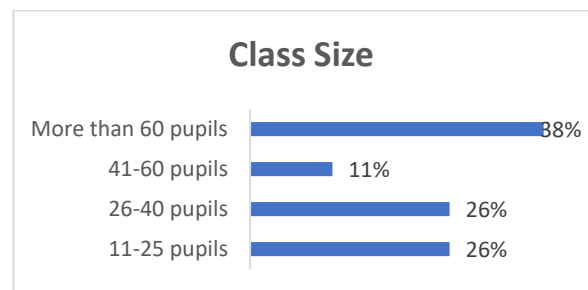


Figure 5.6: Average Class Size

Results emanating from the data, show that majority of the teachers in the sampled schools are female, coming to 73% female teachers, compared to 27% male teachers. On the qualification of individual teachers, it is obvious that a teeming majority of them (over 80%) have the National Certificate of Education, (NCE) which is the minimum standard of certification expected of teachers at the primary school level as their highest educational qualification, this was followed by teachers with Bachelors of Education (B.Ed.) and Bachelors of Science (B.Sc.) graduate teachers. Other information gathered from teacher interviews are the employment status of teachers. Expectedly, majority of the teachers (89% responses) reported that their employment is full time in nature. This contrasts with 8% part-time and 3% volunteers. The experience of teachers also matters a lot for content delivery. Responses from questions on teacher's experience show that 79% teachers have work experience of more than 10 years, this is commendable as only 5% teachers with work experience of 7 – 10 years were reported in the study. Apart from teacher related variables, Teacher-Pupil ratio is a good indicator of teacher-effectiveness and learning outputs. Congestion of classrooms may lead to ineffective class control as over 38% of schools visited during the study reported enrollment of learners at above 60 in a classroom.

6 RESULT OF HEADTEACHER'S INTERVIEW

Head teachers are responsible for administrative management of schools, but most times, they also teach in the school and play other roles as secretaries to the School Based Management Committees (SBMC's) with each of these activities contributing significantly to their effectiveness in the schools. Headteacher questionnaires were therefore administered in the schools visited during the EGMA pilot exercise. The instrument for this category of respondents covers questions on personal attributes as well as school related information.

Data from headteacher interviews show that 89 percent of the schools visited have Pre-Primary/ECCD classes fully established in line with the ECCD policy. However, on school library availability in the visited schools, it is obvious that majority of the schools, accounting for 53 percent reported not to have a dedicated school library. The same situation was observed for the connection of schools to electricity supply and access to potable drinking water where in both cases majority of the schools respond to have none.

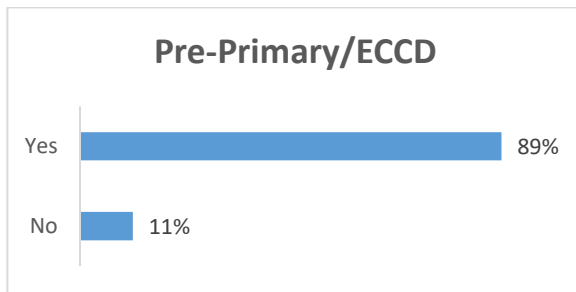


Figure 6.1: Percentage of School with ECCD

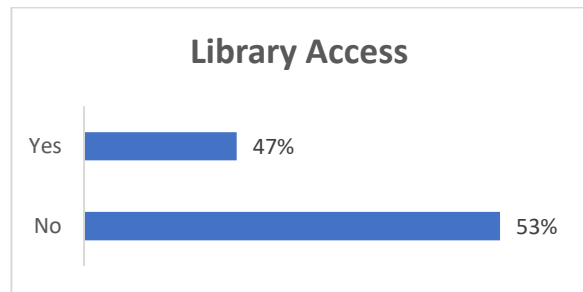


Figure 6.2: School with Library

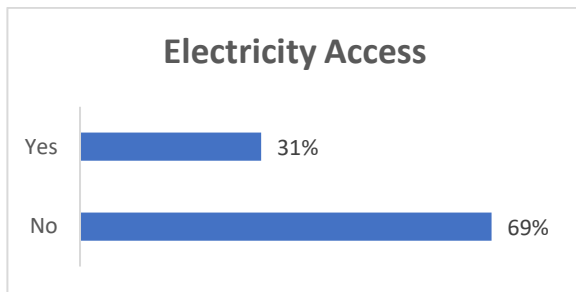


Figure 6.3: School with Electricity

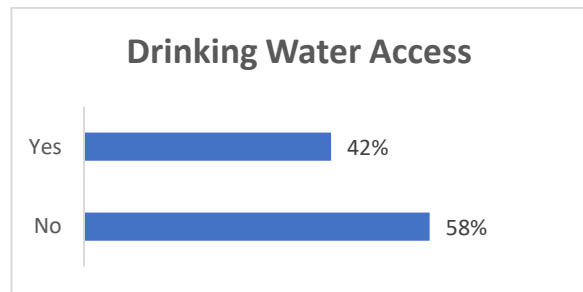


Figure 6.4: Schools with Drinking Water

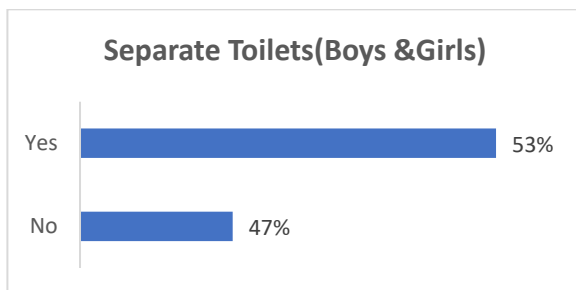


Figure 6.5: Schools with Separate Toilets

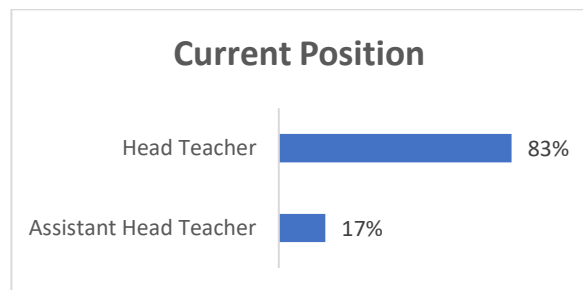


Figure 6.6: Position of Respondent

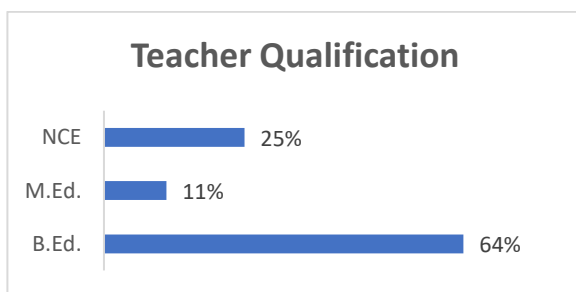


Figure 6.7: Qualification of Respondent

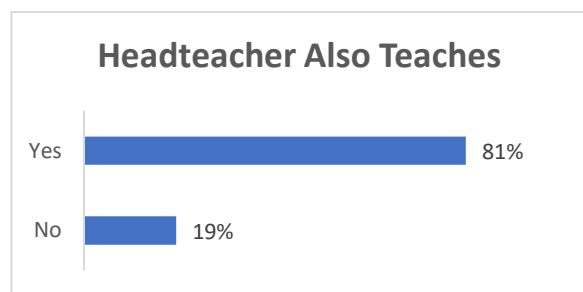


Figure 6.8: Does Head of School also Teach a Class

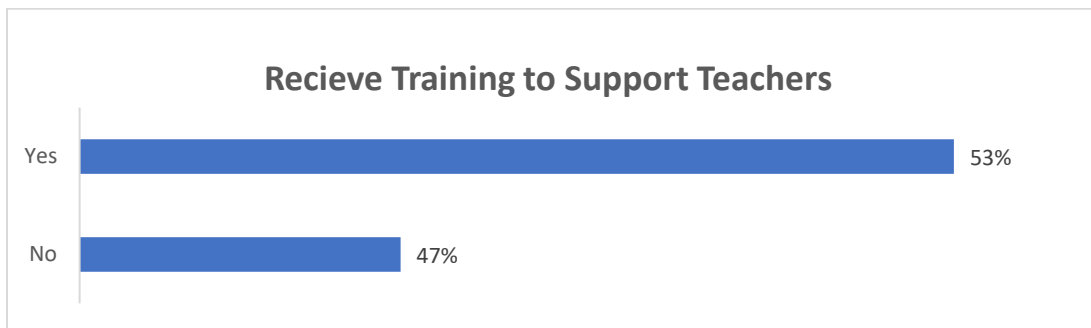


Figure 6.9: Respondent Receive Training to Support

Data on the qualification of headteachers indicate that majority of the headteachers have B.Ed. as their highest educational qualification and are reported to also participate in teaching in their respective schools. Headteachers are expected to have received some training on how to provide support to teachers in their schools. Data emanating from this quality assurance aspect show that a significant percentage (47%) of the headteachers in the sampled schools indicated that they have not received any form of specialized training on how to support teachers.

Lessons Learnt

The results from this study have provided early insight on key aspect of mathematics in early childhood education in Nigeria. It should be noted that this result cannot be used to generalize on the overall situation due to the small sample size and limited coverage. The result from this pilot study will be used to improve the techniques and instruments to be used for a more comprehensive study.

1. Comparison of the results across subtasks showed that the learners performed best in the number identification subtask which is the easiest. 26% of learners in P2 scored above 16 while 34% of learners in P3 scored above 16. The overall score in the subtask is 17.3 for P2 and 23.5 for P3
2. In the Number discrimination subtask, the performance of the learners is slightly above the average score of 5. The learners scored 5.4 and 6.6 out of 10 points in P2 and P3 respectively.
3. The result of the missing number identification subtask indicates a general below average performance of the learners. The average score in this subtask is 2.9 out of 10 for P2 and 3.7 out of 10 for P3 learners.
4. Results from the addition and subtraction subtasks showed learners also performed below average. With most (36% for P2 and 33% for P3 in the addition subtask and 34% for P3 and 37% for P4 in the subtraction subtask) scoring in the range of 1-10. In the addition subtask, learners had an average score of 6.7 in P2 and 8.5 in P3. While in the subtraction subtask, the average score is 4.4 and 5.7 for P2 and P3 learners respectively.
5. The highest number of score (20% for P2 and 23% for P3) in the arithmetic story subtask is in the range of 1-3. This subtask has the lowest performance

with 18% of P2 and 14% of P3 learners scoring zero. The overall average score in this subtask is 1.9 for P2 and 2.3 for P3.

6. 49% of the classes have a population of over 40 learners per class. which is over the recommended for early childhood education. The higher population does not enable teachers to provide adequate attention to the learners and is a contributor to high learner absenteeism observed in the study.
7. 81% of the headteachers reported to also teach learners which is not in accordance with their role as administrators of the school but could be because of inadequate numbers of teachers in the school. Only 53% of the headteachers indicated to have received any kind of training to support teachers.

8 RECOMMENDATIONS AND CONCLUSIONS

Based on the data available from the pilot study, certain observations and inferences indicate a need for certain adjustments to the instruments used for pupils' interviews and a few other instruments used for the pilot study. For instance, in future EGMA studies, it is recommended that additional socioeconomic status/data of the learner's household be obtained to include educational status of parent or caregiver, occupation of parents and caregivers. This will further establish the socio-economic status and its influence on learning. Data should also be segregated into rural and urban locations so that the future interventions can be properly focused where it is needed most. The education status of the learner's parent/guardian/caregiver remains an important factor for consideration as educated parents/caregivers may encourage and assist learners' study for home assignments as part of community participation in the learning process. To further examine the impact of the ECCD (or Nursery/Kindergarten classes) on P2 and P3 learners, it might be necessary to separate learners who passed through the ECCD classes from those who enrolled at P1 directly from home.

The pilot EGMA study has provided additional insights into the importance of mathematics and its in early childhood education. The experience gained from this study also provides an opportunity to review the instruments to be used in the broader and more comprehensive study to accommodate some of the observations above and also address the assumed shortcomings. Overall, the study was a good pilot and it is recommended to be scaled up across the Country.

as soon as the FMoE/MLA is able to secure funding for the National EGMA in line with the recommendations of the National Council on Education in Nigeria.

9 APPENDIX I:

AVERAGE SCORE IN VARIOUS SUBTASKS BY STATE AND GRADE

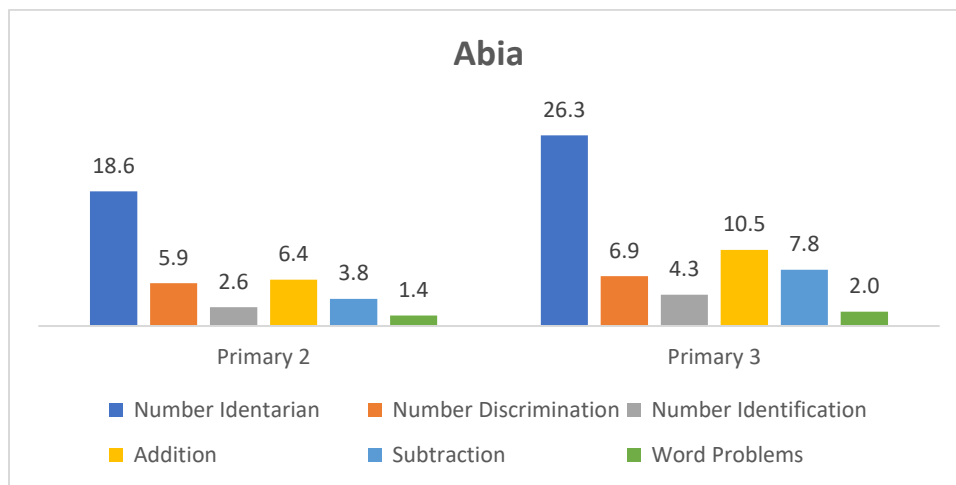


Figure 9.1: Abia State average result in each subtask

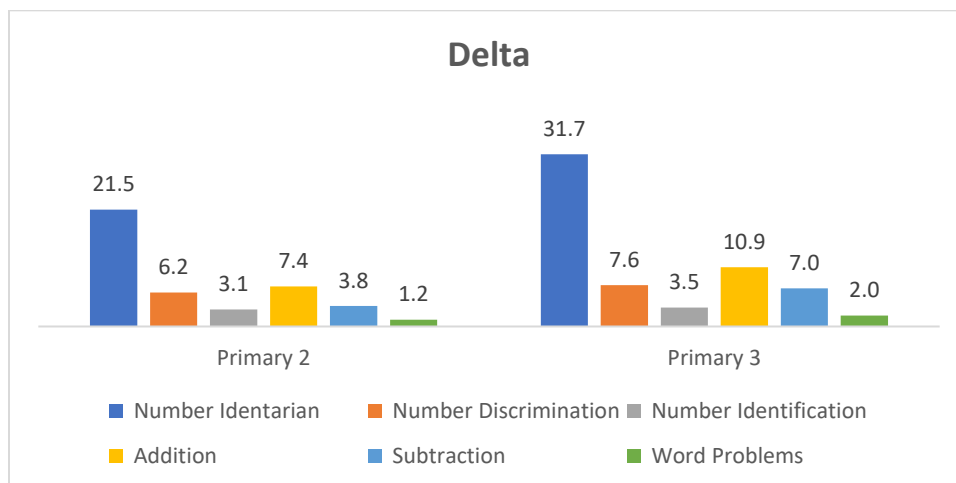


Figure 9.2: Delta State average result in each subtask

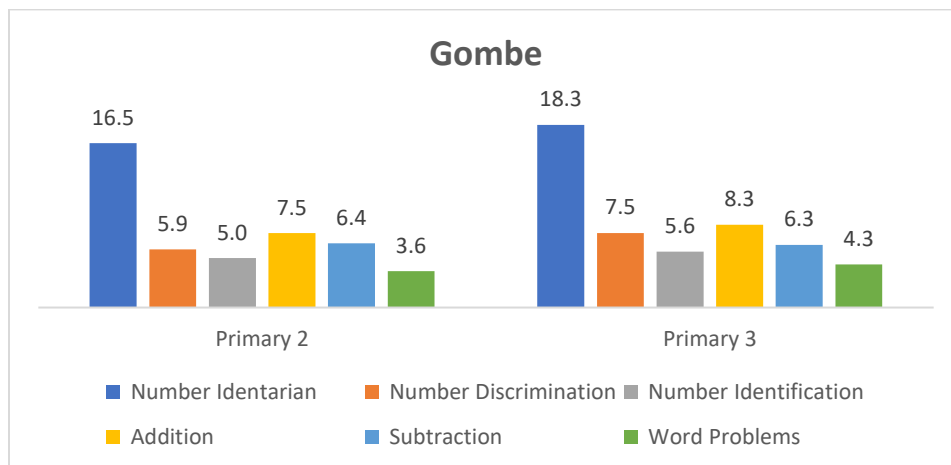


Figure 9.3: Gombe State result in each subtask

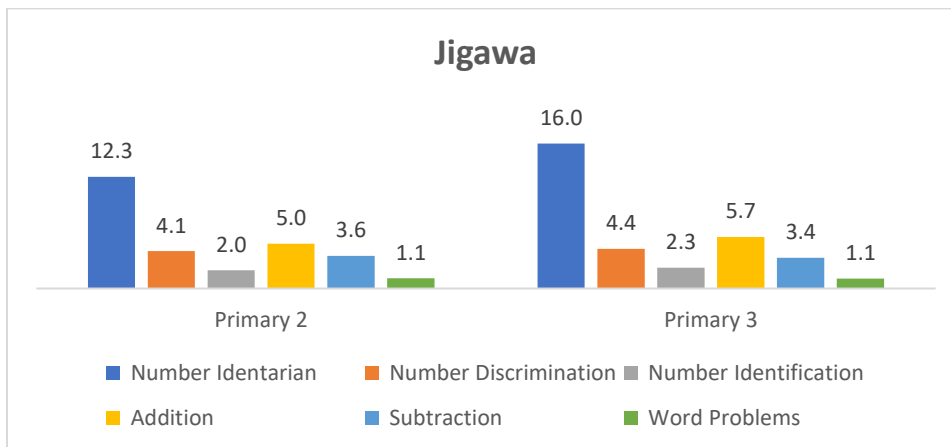


Figure 9.4: Jigawa State result in each subtask

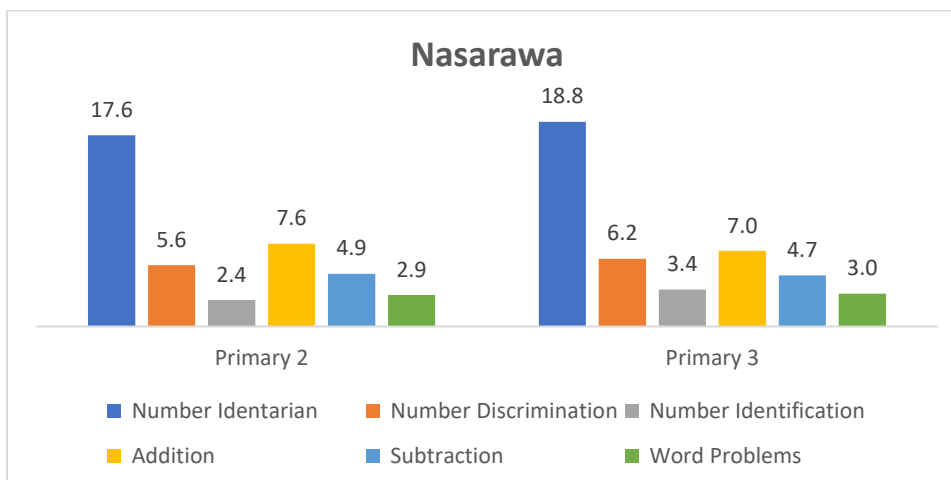


Figure 9.5: Nasarawa State result in each subtask

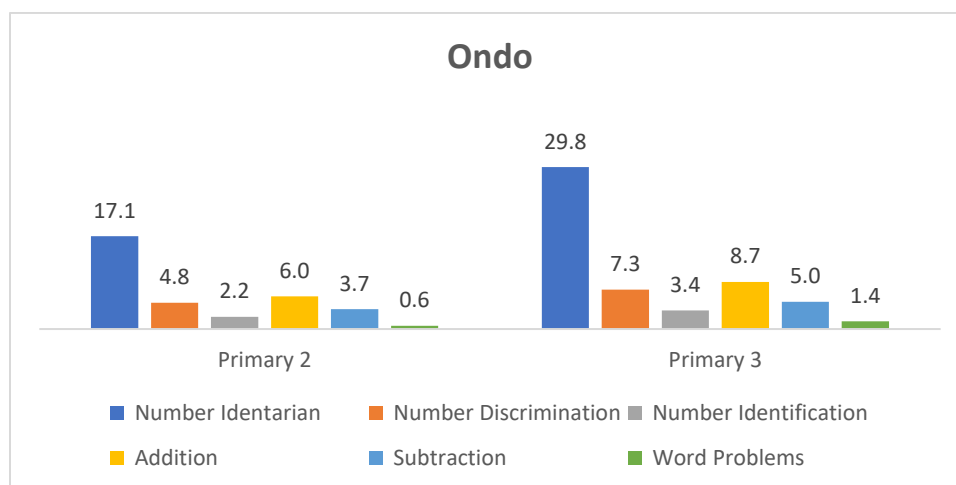


Figure 9.6: Ondo State result in each subtask

10 APPENDIX II:

LEARNER DEMOGRAPHY BY GRADE

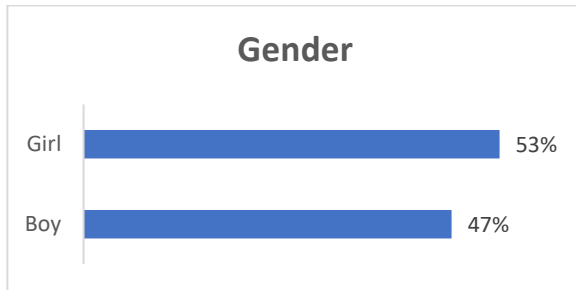


Figure 11.1: Learner Gender Characteristic

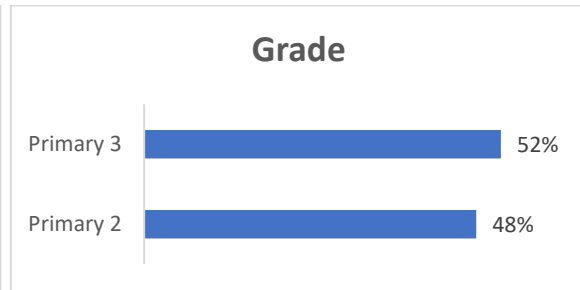


Figure 11.2: Learner's Grade

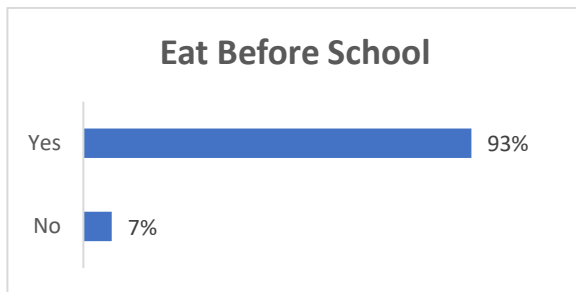


Figure 11.3: Do Learner Eat Before School

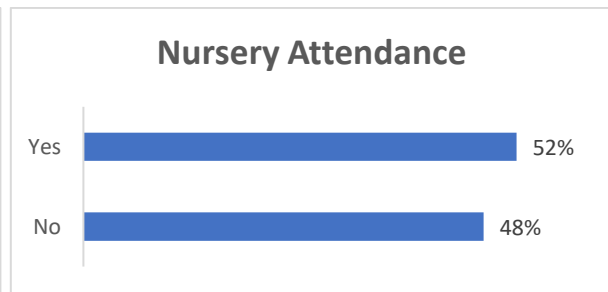


Figure 11.4: Did learner attend Nursery School

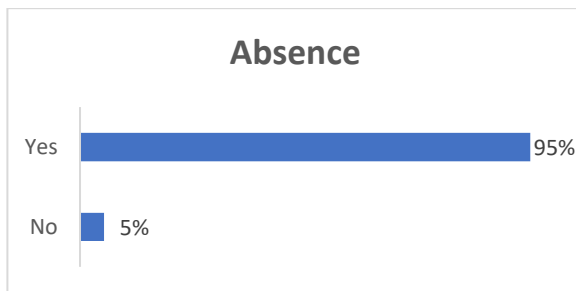


Figure 11.5: Do learner become absent from school

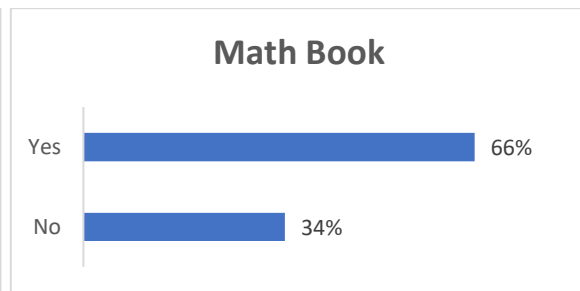


Figure 11.6: Do learner own a math book

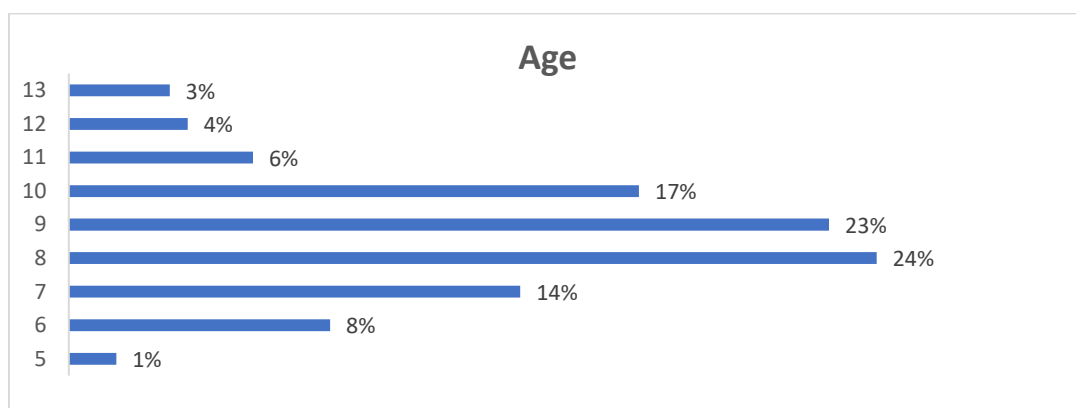


Figure 11.7: Age distribution of learners

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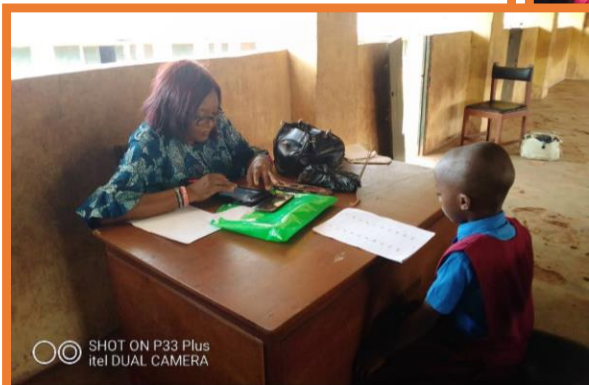
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Picture Gallery (Field Work)

EGMA Assessors and Learners



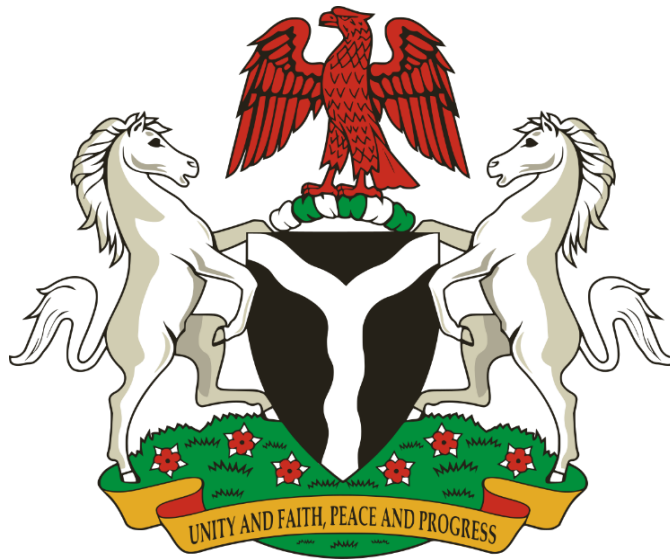
Cross Section of Assessors with Learners and Teachers



WorkShop and Training of EGMA Assessors







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